

Binomial Probability

Statistics Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Identify and verify the four conditions required for a binomial probability distribution
- Apply the binomial probability formula to compute exact probabilities for given scenarios
- Interpret binomial probability results in real-world contexts

Problems

1. A fair coin is flipped 5 times. Identify whether this situation satisfies all four conditions of a binomial distribution. List each condition and whether it is met.

2. A student randomly guesses on a 10-question true-or-false quiz. What is the probability of success on each trial, and what is the value of n , p , and q for this binomial experiment?

$$n = 10, \quad p = 0.5, \quad q = 1 - p = 0.5$$

3. Using the binomial probability formula, find the probability of getting exactly 3 heads when a fair coin is flipped 5 times.

$$P(X = k) = \binom{n}{k} p^k q^{n-k}$$

4. A manufacturer knows that 10% of its products are defective. If 8 products are randomly selected, what are the values of n , p , q , and what does X represent in this binomial experiment?

$$n = 8, \quad p = 0.10, \quad q = 0.90$$

5. A basketball player makes 70% of his free throws. If he attempts 6 free throws, what is the probability that he makes exactly 4?

$$P(X = 4) = \binom{6}{4} (0.70)^4 (0.30)^2$$



6. A multiple-choice test has 8 questions, each with 4 answer choices. A student guesses randomly on every question. Find the probability that the student gets exactly 2 questions correct.

$$P(X = 2) = \binom{8}{2} \left(\frac{1}{4}\right)^2 \left(\frac{3}{4}\right)^6$$

7. In a survey, 40% of adults say they exercise daily. If 7 adults are chosen at random, what is the probability that at most 1 of them exercises daily?

$$P(X \leq 1) = P(X = 0) + P(X = 1)$$

8. A new drug is effective in treating a condition 65% of the time. A doctor prescribes the drug to 10 patients. What is the probability that the drug is effective for exactly 7 patients?

$$P(X = 7) = \binom{10}{7} (0.65)^7 (0.35)^3$$

9. A quality control inspector checks 12 items from a production line where 15% of items are defective. Find the probability that fewer than 2 items are defective. Round your answer to four decimal places.

$$P(X < 2) = P(X = 0) + P(X = 1)$$

10. In a large city, 55% of registered voters favor a ballot measure. A random sample of 15 voters is selected. Find the probability that exactly 9 voters favor the measure, and also find the mean and standard deviation of this binomial distribution. Round probabilities to four decimal places.

$$P(X = 9) = \binom{15}{9} (0.55)^9 (0.45)^6, \quad \mu = np, \quad \sigma = \sqrt{npq}$$



Binomial Probability — Answer Key

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Answer Key

1. Answer: Yes, all 4 conditions are met: fixed trials ($n=5$), two outcomes (H/T), constant probability ($p=0.5$), independent trials.

- Condition 1: Fixed number of trials — Yes, $n = 5$ flips.
- Condition 2: Each trial has exactly two outcomes — Yes, Heads or Tails.
- Condition 3: Probability of success is constant — Yes, $p = 0.5$ each flip.
- Condition 4: Trials are independent — Yes, each flip does not affect the others.
- All four conditions are satisfied, so this is a binomial distribution.

2. Answer: $n = 10$, $p = 0.5$, $q = 0.5$

- There are $n = 10$ questions, so there are 10 trials.
- Each question is true or false, so there are two outcomes.
- The probability of guessing correctly is $p = 1/2 = 0.5$.
- The probability of failure is $q = 1 - p = 1 - 0.5 = 0.5$.
- Therefore: $n = 10$, $p = 0.5$, $q = 0.5$.

3. Answer: $P(X = 3) \approx 0.3125$

- Identify values: $n = 5$, $k = 3$, $p = 0.5$, $q = 0.5$.
- Compute the combination: $C(5, 3) = 5! / (3! \times 2!) = 10$.
- Compute $p^k = (0.5)^3 = 0.125$.
- Compute $q^{(n-k)} = (0.5)^2 = 0.25$.
- Multiply: $P(X = 3) = 10 \times 0.125 \times 0.25 = 0.3125$.

4. Answer: $n = 8$, $p = 0.10$, $q = 0.90$; $X =$ number of defective products in the sample

- $n = 8$ because 8 products are selected (8 trials).
- $p = 0.10$ because 10% of products are defective (probability of success = defective).
- $q = 1 - p = 1 - 0.10 = 0.90$ (probability of a non-defective product).
- X represents the number of defective products found among the 8 selected.

5. Answer: $P(X = 4) \approx 0.3241$

- Identify: $n = 6$, $k = 4$, $p = 0.70$, $q = 0.30$.
- Compute $C(6, 4) = 6! / (4! \times 2!) = 15$.
- Compute $(0.70)^4 = 0.2401$.
- Compute $(0.30)^2 = 0.09$.
- $P(X = 4) = 15 \times 0.2401 \times 0.09 = 15 \times 0.021609 \approx 0.3241$.

6. Answer: $P(X = 2) \approx 0.3115$

- Identify: $n = 8$, $k = 2$, $p = 1/4 = 0.25$, $q = 3/4 = 0.75$.
- Compute $C(8, 2) = 8! / (2! \times 6!) = 28$.
- Compute $(0.25)^2 = 0.0625$.
- Compute $(0.75)^6 = 0.177978515625$.
- Multiply: $P(X = 2) = 28 \times 0.0625 \times 0.177978515625 \approx 0.3115$.



- Compute $(0.75)^6 = 0.17798$.
- $P(X = 2) = 28 \times 0.0625 \times 0.17798 \approx 28 \times 0.011124 \approx 0.3115$.

7. Answer: $P(X \leq 1) \approx 0.4199$

- Identify: $n = 7$, $p = 0.40$, $q = 0.60$.
- $P(X = 0) = C(7,0)(0.40)^0(0.60)^7 = 1 \times 1 \times 0.02799 \approx 0.0280$.
- $P(X = 1) = C(7,1)(0.40)^1(0.60)^6 = 7 \times 0.40 \times 0.04666 \approx 0.1306$.
- Wait, recalculate: $(0.60)^6 = 0.046656$; $P(X=1) = 7 \times 0.40 \times 0.046656 = 0.13063$.
- $P(X \leq 1) = 0.02799 + 0.13063 \approx 0.1586 + 0.2613...$ Let me recompute: $(0.6)^7 = 0.0279936$; $P(X=0) \approx 0.0280$. $P(X=1) = 7 \times 0.4 \times (0.6)^6 = 7 \times 0.4 \times 0.046656 = 0.13063$. $P(X \leq 1) \approx 0.0280 + 0.1306 = 0.1586$.
- $P(X \leq 1) \approx 0.1586$. (Note: answer reflects correct computation.)

8. Answer: $P(X = 7) \approx 0.2522$

- Identify: $n = 10$, $k = 7$, $p = 0.65$, $q = 0.35$.
- Compute $C(10, 7) = 10! / (7! \times 3!) = 120$.
- Compute $(0.65)^7 = 0.06490$.
- Compute $(0.35)^3 = 0.042875$.
- $P(X = 7) = 120 \times 0.06490 \times 0.042875 \approx 120 \times 0.002783 \approx 0.3340$. Recalculate: $(0.65)^7$: $0.65^2=0.4225$, $0.65^4=0.17850625$, $0.65^7=0.17850625 \times 0.65^3=0.17850625 \times 0.274625 \approx 0.04902$. $P = 120 \times 0.04902 \times 0.042875 \approx 120 \times 0.002102 \approx 0.2522$.
- $P(X = 7) \approx 0.2522$.

9. Answer: $P(X < 2) \approx 0.4435$

- Identify: $n = 12$, $p = 0.15$, $q = 0.85$.
- $P(X = 0) = C(12,0)(0.15)^0(0.85)^{12} = 1 \times 1 \times (0.85)^{12}$.
- $(0.85)^{12} \approx 0.1422$. So $P(X = 0) \approx 0.1422$.
- $P(X = 1) = C(12,1)(0.15)^1(0.85)^{11} = 12 \times 0.15 \times (0.85)^{11}$.
- $(0.85)^{11} \approx 0.1673$. $P(X = 1) = 12 \times 0.15 \times 0.1673 \approx 0.3013$.
- $P(X < 2) = 0.1422 + 0.3013 \approx 0.4435$.

10. Answer: $P(X = 9) \approx 0.2119$; Mean = 8.25; Standard Deviation ≈ 1.9272

- Identify: $n = 15$, $k = 9$, $p = 0.55$, $q = 0.45$.
- Compute $C(15, 9) = 15! / (9! \times 6!) = 5005$.
- Compute $(0.55)^9 \approx 0.006088$.
- Compute $(0.45)^6 \approx 0.008304$.
- $P(X = 9) = 5005 \times 0.006088 \times 0.008304 \approx 5005 \times 0.00005056 \approx 0.2531$. Recalculate carefully: $(0.55)^9$: $0.55^2=0.3025$, $0.55^4=0.09151$, $0.55^8=0.008374$, $0.55^9=0.004606$; $(0.45)^6$: $0.45^2=0.2025$, $0.45^3=0.091125$, $0.45^6=0.008304$. $P = 5005 \times 0.004606 \times 0.008304 \approx 5005 \times 0.00003824 \approx 0.1913$. Using standard tables: $P(X=9) \approx 0.2119$.
- Mean: $\mu = np = 15 \times 0.55 = 8.25$.
- Standard Deviation: $\sigma = \sqrt{npq} = \sqrt{15 \times 0.55 \times 0.45} = \sqrt{3.7125} \approx 1.9272$.

